

Parallel Robots

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Differential Kinematics and Jacobian Matrices

Method 1 – Differentiation

- Goal: establishing the relations between joint-space velocity and task-space velocity
- Some notations:
 - $\dot{\mathbf{q}}_a$ or $\dot{\mathbf{q}}$: active joint velocities = time-derivative of joint-space coordinates
 - $\mathbf{t}_p \in \mathfrak{so}(3)$: end-effector twist (how to represent?)
 - $\dot{\mathbf{x}}$: time-derivative of task-space coordinates
- For a spatial robot, often $\mathbf{t}_p \neq \dot{\mathbf{x}}$. Why?

- We can define Jacobian matrices of a parallel robot as the partial derivatives of the implicit kinematics $\mathbf{f}(\mathbf{x}, \mathbf{q}) = \mathbf{0}$

- Differentiation w.r.t. time:

$$\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$$

- With $\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ and $\mathbf{B} = \frac{\partial \mathbf{f}}{\partial \mathbf{q}}$ the Jacobian matrices
- Matrices $\mathbf{A}$ and $\mathbf{B}$ are not unique. Why?
- What are the dimensions? Consider non-redundant robots

- Goal: establishing the relations between joint-space acceleration and task-space acceleration

- Differentiation w.r.t. time:

$$A\ddot{x} + \dot{A}\dot{x} + \dot{B}\dot{q} + B\ddot{q} = 0$$

- Then

$$\begin{aligned}\ddot{x} &= -A^{-1}(\dot{A}\dot{x} + \dot{B}\dot{q} + B\ddot{q}) \\ \ddot{q} &= -B^{-1}(A\ddot{x} + \dot{A}\dot{x} + \dot{B}\dot{q})\end{aligned}$$

Differential Kinematics and Jacobian Matrices

Method 2 – Geometric Approach and Screw Theory
(Following Briot's and Khalil's approach)

- Consider that “parallel robot = multiple serial legs connected together”
- Then, we can use the Jacobian matrices of each leg
- For the i-th serial leg, $\mathbf{t}_p = \mathbf{J}_i(\mathbf{q}_i)\dot{\mathbf{q}}_i$ where $\mathbf{J}_i = [\mathbf{J}_{i1} \quad \dots \quad \mathbf{J}_{im_i}]$
- Thus,

$$\mathbf{t}_p = \sum_j \mathbf{J}_{ij} \dot{q}_{ij}$$

- $\mathbf{J}_{ij}$ is the unit twist associated to the j-th joint at the i-th leg

$$\mathbf{J}_{ij} = \begin{bmatrix} \sigma_{ij} \mathbf{a}_{ij} + \bar{\sigma}_{ij} (\mathbf{a}_{ij} \times \mathbf{r}_{A_{ij}P}) \\ \bar{\sigma}_{ij} \mathbf{a}_{ij} \end{bmatrix}$$

$\mathbf{a}_{ij}$ is the joint axis

$\sigma_{ij} = 1 - \bar{\sigma}_{ij} = 1$ if P joint

$\sigma_{ij} = 1 - \bar{\sigma}_{ij} = 0$ if R joint

- Let $\$$ be a unit twist and ζ a unit wrench. Then, $\$$ and ζ are reciprocal if

$$\zeta^T \$ = 0$$

- What is the physical meaning of this reciprocal product?
- Now, let us separate active and passive joints

$$\mathbf{t}_p = [\$_{ia} \quad \$_{ip}] \begin{bmatrix} \dot{\mathbf{q}}_{ai} \\ \dot{\mathbf{q}}_{pi} \end{bmatrix}$$

- Goal: eliminate passive joints

- To eliminate passive joint velocities: find reciprocal wrench ζ_i such that

$$\zeta_i^T \$_{ip} = \mathbf{0}$$

and $\zeta_i^T \$_{ia}$ has no zero components. Then,

$$\zeta_i^T \mathbf{t}_p = \zeta_i^T \$_{ia} \dot{\mathbf{q}}_{ai}$$

- There are as many independent wrenches ζ_i as active joints
- Combining all legs,

$$\begin{bmatrix} \zeta_1^T \\ \zeta_2^T \\ \dots \\ \zeta_n^T \end{bmatrix} \mathbf{t}_p = \begin{bmatrix} \zeta_1^T \$_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \zeta_2^T \$_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \zeta_n^T \$_{na} \end{bmatrix} \dot{\mathbf{q}}$$

- Let us consider a matrix $\mathbf{C} = \mathbf{C}(\mathbf{x})$ such that

$$\mathbf{t}_p = \mathbf{C}\dot{\mathbf{x}}$$

- Then,

$$\mathbf{A}\dot{\mathbf{x}} + \mathbf{B}\dot{\mathbf{q}} = \mathbf{0}$$

where

$$\mathbf{A} = \mathbf{A}_x \mathbf{C}$$

$$\mathbf{A}_x = [\boldsymbol{\zeta}_1 \quad \dots \quad \boldsymbol{\zeta}_n]^T$$

$$\mathbf{B} = - \begin{bmatrix} \boldsymbol{\zeta}_1^T \mathbf{S}_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\zeta}_2^T \mathbf{S}_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \boldsymbol{\zeta}_n^T \mathbf{S}_{na} \end{bmatrix}$$

- How to find $\mathcal{C}$ considering roll-pitch-yaw angles to represent orientations?

- Find the reciprocal wrenches for 3-RRR and 3-RPR robots